

RAMANJEET SAINI

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TECHNICAL SKILLS

GenAI & LLMs: LangChain, LangGraph, LlamaIndex, RAG, multi-agent systems, GPT-4o, Claude (AWS Bedrock), Llama 3, QLoRA, vLLM, RAGAS, DeepEval, LangSmith | **ML / DL:** PyTorch, scikit-learn, XGBoost, Hugging Face Transformers | **MLOps:** AWS SageMaker, MLflow, W&B, Airflow, Docker, Evidently AI | **Data & Vector DBs:** FAISS, OpenSearch, Redis, PostgreSQL, DynamoDB, AWS S3 | **Languages:** Python (expert), SQL · FastAPI, REST · GitLab CI/CD | **Domains:** LLM fine-tuning & alignment, agentic workflows, RAG pipelines, document AI, fraud & risk modeling, recommendation systems

EDUCATION

The University of Texas at Austin

May 2023

M.S., Information Technology & Management | Concentrations: Machine Learning, Data Science | GPA: 3.96 / 4.0

Pandit Deendayal Energy University

May 2018

B.Tech., Mechanical Engineering

EXPERIENCE

Western Union

Austin, TX

AI / Machine Learning Engineer

Aug 2023 – Present

- Cut fraud-analyst handle time by **26%**, reduced false-positive rate by **14%**, and raised straight-through processing by **19 pp** by productionizing a **RAG pipeline** over OpenSearch with hybrid dense/sparse retrieval using **LlamaIndex + Claude on AWS Bedrock**; evaluated with **RAGAS** and traced with **LangSmith**.
- Protected a **35 bps** margin target and lifted checkout conversion by **9%** by designing a production **multi-agent pricing platform** using **LangGraph + Claude 3 (AWS Bedrock)**; enforced AI governance via Bedrock Guardrails and AWS KMS; eliminated 80% of manual analyst effort.
- Saved **\$0.9M/quarter** in manual review costs by shipping a KYC identity-verification API at **97% precision / 94% recall** and **220 ms p95** using **Claude (Bedrock) + Amazon Textract**; served via FastAPI / API Gateway / Lambda with full GDPR / AML compliance.
- Outperformed baseline by **14 F1 points** by fine-tuning **Llama 3.1 8B with QLoRA (PEFT)** and aligning via **DPO (TRL)**; served via **vLLM with AWQ 4-bit quantization**, cutting GPU memory 55% with <2% accuracy drop.
- Drove **\$1.1M/quarter** incremental revenue by architecting a **two-tower retrieval and ranking model (PyTorch + FAISS)** deployed on **Triton Inference Server with ONNX** at <80 ms p95; validated via A/B experiment.
- Lifted customer retention by **12 pp** and cut model release cycle by **30%** by owning end-to-end MLOps for churn and CLV models on **SageMaker + Kubeflow Pipelines** with MLflow, W&B, and Evidently AI drift monitoring.

Infosys Limited

Pune, India

Data Engineer

Jul 2020 – Jun 2022

- Cut infrastructure costs by **34%** by managing **Apache Spark / AWS EMR** batch pipelines processing 500M+ records/week with ML-based anomaly detection and **AWS Kinesis** real-time streaming.
- Improved ML personalization pipeline throughput by **12%** via **PySpark + Kafka → S3 → DynamoDB** pipelines; cut serverless costs by **33%** by optimizing AWS Lambda inference functions.

Cloud Engineer

Oct 2018 – Jul 2020

- Reduced DBaaS (PostgreSQL, SQL Server, Oracle) downtime by **28%** via CloudWatch monitoring; deployed Tableau connectors via GitLab CI/CD, improving data reliability by **18%**.

PROJECTS

AI Music Composition Assistant (LangGraph + Claude) — github.com/ramanjeetsaini

2025

- Built a personal multi-agent music co-writer using **LangGraph + Claude (Bedrock)** with specialized agents for chord progression generation, lyric suggestion, and melody structuring based on user mood and genre input; used **DSPy** for structured musical outputs and **LangSmith** for end-to-end tracing; surface via a Streamlit interface.

Snap & Track: Vision-Based Calorie Estimator (GPT-4o Vision + QLoRA) — github.com/ramanjeetsaini

2025

- Built a personal meal logging app that identifies food items and estimates calories from a phone photo using **GPT-4o Vision**; fine-tuned **Llama 3.1 8B with QLoRA + DPO** on a custom food-description dataset to improve portion size reasoning; served via **vLLM (AWQ 4-bit)** cutting GPU memory 55%; evaluated nutritional accuracy with DeepEval.

Flight Deal Finder: Cash vs. Miles Optimizer (RAG + Agentic AI) — github.com/ramanjeetsaini

2024

- Built a personal travel tool that compares flight options by cash price and miles redemption value using **LangChain + GPT-4o**; scraped and embedded live fare and award data into **FAISS** (<50 ms retrieval); added a cross-encoder re-ranking stage to surface best-value routes; achieved **>0.85 RAGAS answer-relevance score** with a Streamlit UI.